1 INPUT / HOT-SWAP

J1 XT90PW-M battery

D1 SMCJ58A

C1 1uF 100V

C59 33uF 100V

C60 47uF 100V

RS1 1.8m 1% 3W

RS2 1.8m 1% 3W

Q1 CSD19536KTT

Q2 CSD19536KTT

R3 10R

R4 10R

Q7 BSC040N10NS5ATMA1

Q8 BSC040N10NS5ATMA1

R33 10R

R34 10R

Green = PGD released: not a bus-voltage test

RS1 || RS2 = 0.9 mR (limit 53-69 A). Q1 || Q2 share the series path.

Q7 || Q8 + U9 = ideal diode: regen cannot reach the source

U9 LTC4359IMS8#PBF

D7 BZT52C12

D8 SMAT70A

R35 1k

R10 1k

C3 47nF 100V

R11 100k

R12 1k

D5 green PGD

Q5 BSS138

U1 LM5069M-2/NOPB

R5 71.5k 1%

R6 10.7k 1%

R7 71.5k 1%

R8 3.16k 1%

R9 27.4k 1%

C2 15nF

Q4 BSS138

UV on/off: 19.2 / 17.7 V

OV off/on: 59.1 / 57.6 V

2 DC LINK / CHAIN PORTS

12 x 130 uF = 1560 uF nominal

C20 130uF 80V

C21 130uF 80V

C22 130uF 80V

C23 130uF 80V

C24 130uF 80V

C25 130uF 80V

C26 130uF 80V

C27 130uF 80V

C28 130uF 80V

C29 130uF 80V

C30 130uF 80V

C31 130uF 80V

C32 1uF 100V

C33 1uF 100V

C34 1uF 100V

C35 1uF 100V

R13 10k 0.5W

R14 10k 0.5W

J2 XT30PW(2+2)-M chain

J3 XT30PW(2+2)-M chain

JP1 CAN TERM

R15 60.4R

R16 60.4R

C36 4.7nF

JP1 closed: endpoint termination

JP1 open: through node

3 HOUSEKEEPING: 12 V / 3.3 V

D3 SS1H10

D4 SS1H10

C52 2.2uF 16V

C50 2.2uF 100V

C51 0.1uF 100V

R40 25.5k 1%

C53 10nF

U7 LT8631EFE#PBF

L1 22uH

C54 0.1uF 25V

R41 1M 1%

C55 4.7pF

C56 22uF 25V

R42 71.5k 1%

D4 sustains control power from the charged bus after battery removal.

U8 MCP1703AT-3302E/TT

C57 1uF 25V

C58 1uF

3.3 V supplies the comparator, reference and interlock logic.

4 REGENERATION / BRAKE COMMAND

Targets: 57.1 V ON / 55.0 V OFF. CH_CMD = CH_CMP OR DISCH. Driver: CH_CMD OR backup (block 7).

R20 150k 0.1%

R21 150k 0.1%

R22 6.81k 0.1%

R24 237k 0.1%

R23 237k 0.1%

U3 TLV3201AIDBVR

U4 REF3312AIDBZR

C41 1uF

C40 0.1uF

C42 1uF

U5 SN74LVC1G32DBVR

C43 0.1uF

D14 1N5819HW

D13 1N5819HW

U6 UCC27517DBVR

C44 0.1uF 25V

C45 1uF 25V

R25 4.7R

R26 10k

R27 1k

D6 red BRAKE

J4 BRAKE RESISTOR

D2 V10P10-M3/86A

Q3 BSC040N10NS5ATMA1

7 BACKUP OVERVOLTAGE BRAKE: BUS-POWERED, NO AUX

U10 LR8N8-G

C61 0.1uF 100V

R43 2.4k 1%

R44 17.4k 1%

C62 1uF 25V

Fires Q3 through the shared driver above 61.1 V (60.4-61.9 V), off at 60.3 V, whatever the aux does.

Primary chopper 57.1 V; bank 80 V. Driver supply = +12V OR V_CHB (D14/D15).

R45 750k 0.1%

R46 750k 0.1%

R47 9.88k 0.1%

C64 1nF

U11 TPS3700DDCR

C63 0.1uF

R48 10k

D12 1N5819HW

R49 100k

D15 1N5819HW

5 RUN / THERMOSTAT INTERLOCKS

J5

D11 BAT54WS

R30 10k

C46 0.1uF

R31 10k

RUN (close to run)

Q6 BSS138

R32 10k

J6

C47 0.1uF

THERMOSTAT (NC)

J5 open/broken: OFF + dump request. J6 open: inhibit dump only; OV clamp stays active.

6 HOT-SWAP ENABLE: RUN CLOSED AND AUX 12 V GOOD

R36 1M 1%

D9 BZT52C10

C38 1nF

Q9 BSS138

U12 TPS3700DDCR

R50 255k 0.1%

R51 10k 0.1%

R52 150k

R37 100k

C65 0.1uF

C66 0.1uF

AUX_OK released only above 10.6 V on +12V: C65 holds Q9 off through the 450 us TPS3700 start-up.

Design review in progress; routing held. See support-board-design-review.md

Sheet: /

File: SupportBoard.kicad_sch

Title: Actuator support board | power and control

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